

2024, Mar. 2024 Release LOC:Acute Adult

Diabetes Mellitus

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review (1, 2, 3, 4)

Episode Day 1 (1, 3, 4)

Episode Day 2 (4, 26, 27)

Episode Day 3 (35)

Discharge Screens (36)

Utilization Benchmarks

Notes

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Overview

Instruction: The Diabetes Mellitus (DM) subset is for patients with new-onset DM and pre-existing DM who are experiencing hyperglycemia.

- For patients with diabetic ketoacidosis, refer to the Diabetic Ketoacidosis subset
- For patients with a hyperosmolar hyperglycemic state, refer to the Hyperosmolar Hyperglycemic State subset
- For patients with hypoglycemia, refer to the Hypoglycemia subset
- For patients with pregestational DM or gestational DM, refer to the Antepartum subset

Diabetes mellitus (DM) is a complex chronic condition that requires ongoing medical care, multiple interventions, and risk reduction strategies to improve outcomes. Type 1 diabetes, type 2 diabetes, diabetes due to other causes, and gestational diabetes mellitus (GDM) are the four general classification categories of DM. Type 1 DM typically leads to complete deficiency of insulin, is caused by autoimmune beta cell destruction, and can occur in adulthood as latent autoimmune diabetes. Loss of sufficient beta cell insulin secretion causes type 2 DM, typically due to insulin resistance. Chemical-induced and drug-induced DM, exocrine pancreas diseases, and monogenic diabetes syndromes represent types of diabetes that originate from other causes.

The clinical presentation and disease progression of type 1 and type 2 DM can significantly differ. Both type 1 DM and type 2 DM can occur in children and adults. To determine the treatment regimen, the classification of DM is essential. However, at the time of diagnosis, some patients cannot be distinctly classified into one of these categories and misdiagnoses can occur. As time progresses, beta cell deficiency in patients becomes more evident. Multiple genetic and environmental factors cause loss of beta cells or beta cell function, resulting in hyperglycemia. The risk of acute complications such as diabetic emergencies and long-term microvascular and macrovascular complications increases in all people with diabetes once hyperglycemia develops (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

Evaluation and Treatment:

Diagnostic tests for diabetes mellitus (DM) include an A1C value greater than or equal to 6.5% (48 mmol/mol), fasting plasma glucose value greater than or equal to 126mg/dL (7.0mmol/L) measured when there has been no caloric intake for a minimum of eight hours, or a two-hour plasma glucose value greater than or equal to 200 mg/dL (11.1 mmol/L) obtained during a 75g oral glucose tolerance test. A diagnosis of DM can occur when a patient with symptoms of hyperglycemia or hyperglycemic crisis has a random plasma glucose value greater than or equal to 200 mg/dL (11.1 mmol/L). In patients with no symptoms of hyperglycemia, the diagnosis requires two abnormal screening test results

from the same or separate samples. Knowing the plasma glucose level and confirming that symptoms are a result of DM is vital to making informed decisions related to treatment. To improve inpatient management and discharge planning, hospitalized patients with DM should have A1C measured if not evaluated within the previous three months. When possible, consulting and collaborating with providers and teams trained in diabetes management is encouraged to reduce hospital length of stay, reduce readmission following hospitalization, improve outcomes, and enhance glycemic control. Clinical judgment, ongoing assessment, and evaluation of clinical status, such as socioeconomic and psychosocial factors, the severity of illness, comorbidities, nutritional intake, medications that may affect glucose levels, and changes and trends in measured glucose levels, are necessary to make individualized treatment decisions (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾● Diabetes mellitus and, **One:**– Blood glucose > 400 mg/dL(22.2 mmol/L) and new onset diabetes mellitus ⁽⁶⁾● Blood glucose > 500 mg/dL(27.8 mmol/L) and, ≥ **One:**● Creatinine, **Both:** ⁽⁷⁾– ≥ 1.5x to < 3x baseline or estimated baseline ^(8, 9)

– < 4 mg/dL(353.6 μmol/L)

– Heart rate > 100/min, sustained ⁽¹⁰⁾– Inability to tolerate or adhere to prescribed regimen (including diet) ^(11, 12, 13)– Intractable nausea and vomiting ⁽¹⁴⁾– Mental status changes (excludes coma, stupor, or obtundation) ⁽¹⁵⁾– Na > 150 mEq/L(150 mmol/L) ⁽¹⁶⁾● Orthostatic hypotension, **Both:** ⁽¹⁷⁾

– Sustained drop in blood pressure within 3 min of sitting or standing

● Blood pressure, ≥ **One:**

– Systolic drop ≥ 20 mmHg

– Diastolic drop ≥ 10 mmHg

● **Episode Day 1, One:** ^(1, 3, 4)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾● Diabetes mellitus and, **Both:**● Finding, **One:**– Blood glucose > 400 mg/dL(22.2 mmol/L) and new onset diabetes mellitus ⁽⁶⁾● Blood glucose > 500 mg/dL(27.8 mmol/L) and, ≥ **One:**● Creatinine, **Both:** ⁽⁷⁾– ≥ 1.5x to < 3x baseline or estimated baseline ^(8, 9)

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– Sustained drop in blood pressure within 3 min of sitting or standing

● Blood pressure, ≥ **One:**

– Systolic drop ≥ 20 mmHg

– Diastolic drop ≥ 10 mmHg

● Intervention, **Both:** ^(18, 19)– Insulin or oral glucose-lowering medication initiation or adjustment ^(20, 21, 22, 23)– IV fluid ^(24, 25)● **Episode Day 2, One:** ^(4, 26, 27)● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁸⁾– Blood glucose within acceptable limits ^(29, 30)

– Patient or caregiver knowledgeable about disease and understands the regimen

– Tolerating PO ⁽³¹⁾● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**● Blood glucose, **Both:**

● Finding, **One:**● Glucose, $\geq$ **One:** ⁽²¹⁾

- > 250 mg/dL(13.9 mmol/L)
- < 70 mg/dL(3.9 mmol/L)
- Insulin or oral glucose-lowering medication initiation, adjustment, or dose held ^(20, 22, 23)
- Inadequate oral intake and IV fluid ^(24, 25, 32)
- Patient or caregiver knowledge deficit and diabetes education ^(33, 34)

● Episode Day 3, **One:** ⁽³⁵⁾● OBSERVATION, **One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁸⁾

- Blood glucose within acceptable limits ^(29, 30)
- Patient or caregiver knowledgeable about disease and understands the regimen
- Tolerating PO ⁽³¹⁾

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**

- For a condition other than diabetes mellitus use appropriate subset (Episode Day 1) based on clinical findings
- Refer for secondary review

Discharge, Level of Care, One: ⁽³⁶⁾● HOME, **Both:**● Level of care appropriateness, **Both:**

- Home environment safe and accessible ⁽³⁷⁾
- Patient or caregiver demonstrates ability to manage care

● Diabetes mellitus discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁸⁾

● Education on the importance of, **All:**

- Prevention and treatment of hypoglycemia
- Home monitoring of blood glucose
- Sick day management
- Foot inspection and care
- Pneumococcal and annual influenza vaccine
- Annual eye examination
- Lifestyle modification to prevent hypertension and dyslipidemia ⁽³⁹⁾
- Smoking cessation, if indicated
- Glycemic control and hypertension management to prevent retinopathy, nephropathy, and neuropathy ⁽⁴⁰⁾
- Medication reconciliation ⁽⁴¹⁾
- Follow-up care planned ⁽⁴²⁾
- Identify and address transportation needs

● HOME CARE, **Both:**● Level of care appropriateness, **All:**

- Home environment safe and accessible ⁽³⁷⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(43, 44)

● Skilled services, $\geq$ **One:** ⁽⁴⁵⁾

- Skilled nursing ⁽⁴⁶⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁴⁷⁾

● Diabetes mellitus discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁸⁾

● Education on the importance of, **All:**

- Prevention and treatment of hypoglycemia

- Home monitoring of blood glucose
- Sick day management
- Foot inspection and care
- Pneumococcal and annual influenza vaccine
- Annual eye examination
- Lifestyle modification to prevent hypertension and dyslipidemia ⁽³⁹⁾
- Smoking cessation, if indicated
- Glycemic control and hypertension management to prevent retinopathy, nephropathy, and neuropathy ⁽⁴⁰⁾

- Medication reconciliation ⁽⁴¹⁾
- Follow-up care planned ⁽⁴²⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

- **Level of care appropriateness, One:**
 - Support systems available ⁽⁴⁸⁾
- **Skilled treatment, ≥ One:**
 - Clinical assessment ⁽⁴⁹⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁵⁰⁾

● **SKILLED MEDICAL OR THERAPY, Both:**

- **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
- **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁴⁶⁾
- **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(51, 52)
 - Functional impairments requiring at least supervision ⁽⁵³⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

- **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
- **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁴⁶⁾
- **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(51, 52)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵³⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines

- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● **SUBACUTE REHAB, Both:**

- **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level

- **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(51, 52)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵³⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5 d/wk and ≥ 2 therapy disciplines
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - **Level of care appropriateness, Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3 x/wk
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(51, 52)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵³⁾
 - Able to tolerate ≥ 15 h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁵⁴⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
 - **In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵⁵⁾
 - Ventilator dependent ≥ 6 h/d and weaning planned ^(56, 57)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁵⁸⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
637 DIABETES WITH MCC	n/a	4.1	CMS GMLOS
638 DIABETES WITH CC	n/a	3	CMS GMLOS
639 DIABETES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
DIABETES MELLITUS	24%	3.1	InterQual
PREGESTATIONAL DIABETES MELLITUS	25%	3	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Expected progress after 24h

- Glucose level improved
- Vital signs stable and within acceptable limits
- PO intake improving
- Activity increasing

Consider:

- Changing medications to PO or stabilizing insulin regimen for discharge
- Education/training in medication administration for new insulin dependent diabetic
- Blood pressure management for hypertension
- Nutrition consultation and/or diabetic nurse educator
- Discharge planning

4:

Care Facilitation

Potential risk for readmission

- Knowledge deficit or non-adherence with condition management identified (**Home care, Readmission Reduction Program**)
- Knowledge deficit with sick day management

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, in-home services, and transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or Activities of Daily Living (ADL) impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self

- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on disease management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application and information should be given to patient
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment), consider Intensive Case Management intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

Individuals with newly diagnosed diabetes may be unaware of the clinical symptoms of diabetes mellitus, such as polyuria, nocturia, polydipsia, blurred vision, and polyphagia, which can result when significant hyperglycemia surpasses the renal glucose threshold. Increased plasma glucose levels may cause glycosuria, increased urine

osmolality, and subsequent diuresis. Increased diuresis contributes to polyuria and nocturia. Fluid losses stimulate thirst and lead to polydipsia and consequent fluid shifts involving the eye's lens can cause blurred vision. Also, the substantial loss of calories in the urine may cause polyphagia. Lastly, insulin deficiency accelerates protein catabolism and the breakdown of fat, all of which can lead to unintentional weight loss (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Winter et al., In: Winter et al., ed. Handbook of Diagnostic Endocrinology, 3rd edn. 2021:238-9).

7:

Risk for acute kidney injury increases in individuals with diabetes. Impaired kidney function increases the potential for difficult glycemic control and treatment-related hypoglycemia challenges. The presence of chronic kidney disease (CKD) impacts the risks and benefits associated with rigorous glycemic control and many glucose-lowering medications. Adverse effects such as hypoglycemia and mortality are more likely to occur in patients with CKD (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Kidney Disease: Improving Global Outcomes Diabetes Work, Kidney Int 2022, 102: S1-S127).

8:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

9:

When the baseline serum creatinine level is unknown, the Modification of Diet in Renal Disease (MDRD) equation can be used to estimate an expected creatinine value (assuming a normal baseline value for glomerular filtration rate (GFR) of 75ml/min per 1.73 meters squared). The following values demonstrate one method to estimate baseline serum creatinine (eSCr) (with the omission of the race variable) (Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138; Bellomo et al., Crit Care 2004, 8: R204-12):

Male sex assigned at birth:

- Age 20-24 - eSCr 1.3 mg/dL (115 µmol/l)
- Age 25-39 - eSCr 1.2 mg/dL (106 µmol/l)
- Age 40-65 - eSCr 1.1 mg/dL (97 µmol/l)
- Age > 65 - eSCr 1.0 mg/dL (88 µmol/l)

Female sex assigned at birth:

- Age 20-29 - eSCr 1.0 mg/dL (88 µmol/l)
- Age 30-54 - eSCr 0.9 mg/dL (80 µmol/l)
- Age > 54 - eSCr 0.8 mg/dL (71 µmol/l)

Both the MDRD and the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations provide a calculation of estimated GFR using serum creatinine, sex assigned at birth, age, and race. Recognition of substantial diversity within population groups has led to multiple organizations no longer including race in the estimated GFR calculations. This approach was investigated by a joint task force of the National Kidney Foundation and the American Society of Nephrology, the result of which was a recommendation to implement the new CKD-EPI creatinine equation which does not use the race variable (Delgado et al., Am J Kidney Dis 2022, 79: 268-88 e1; Inker et al., N Engl J Med 2021, 385: 1737-49).

10:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

11:

Application of this criteria point is appropriate for patients who cannot tolerate or adhere to their prescribed treatment regimen, including those who fail outpatient treatment.

12:

Social determinants of health can cause significant health inequities related to diabetes and an inability to adhere to the treatment regimen which may contribute to complications and poor outcomes. Financial barriers such as medications and device costs contribute to medication nonadherence and poorly controlled glycemic levels. Food insecurity and the inability to consistently obtain nutritious foods are associated with a higher risk for uncontrolled glycemic levels, decreased medication adherence, sub-optimal self-care behaviors, and depression. Housing insecurity can often coincide with food insecurity, decreased health literacy and health numeracy deficiencies, lack of insurance coverage, mental illness, and cognitive dysfunction. Patients with diabetes and home insecurity require secure places to keep and store diabetes medications and supplies to adhere to their prescribed regimen. Language barriers can contribute to an increased risk of diabetes complications in non-English speakers. Incorporating language assistance and offering education materials in multiple languages can reduce language barriers and improve these patients' communication, knowledge, adherence, and health outcomes. Communication of diabetes care and education should be presented in a manner that is easy to comprehend. Health numeracy requires numeric skills such as interpreting food labels and blood glucose levels for medication administration. Health literacy and numeracy correlate with patients' ability to manage their disease and self-care effectively. They also contribute to a patient's understanding of diabetes management and complication prevention concepts and are essential to establishing and promoting adherence to a treatment regimen (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

13:

Malfunction and failure of diabetes technology can represent barriers that contribute to an inability to tolerate and adhere to the diabetes medical management plan; early recognition and management are essential. Diabetes technology refers to devices, software, and hardware used to enhance diabetes condition management, quality of life, and the health of people with diabetes. The complexity and changing landscape of diabetes technologies requires planning, training, education, and follow-up to prevent complications from these barriers. Additional education needs and potential barriers should be assessed periodically in case of complications and unmet goals and/or outcomes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

14:

Vomiting and pronounced hyperglycemia necessitate adjustment of the treatment regimen and expeditious interaction with a diabetes care team. A patient managed with noninsulin pharmacological agents or medical nutrition only may require insulin in addition to ensuring sufficient fluid and caloric intake (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

15:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

16:

Hyperglycemia can cause variable sodium levels directly related to intracellular and extracellular water distribution caused by compensatory osmotic shifts. Hypernatremia can occur when hyperglycemia causes osmotic diuresis due to glucosuria, leading to volume depletion (Khan et al., Pak J Med Sci 2019, 35: 797-801).

17:

Orthostatic hypotension (OH) is a sustained decrease in systolic blood pressure greater than or equal to 20 mmHg or a decrease in diastolic blood pressure greater than or equal to 10 mmHg within 3 minutes of standing or a head-up tilt (HUT) test (Fedorowski et al., Circ Arrhythm Electrophysiol 2022, 15: e010573). The orthostatic stress tests of active standing or HUT assess for postural changes in blood pressure. However, the HUT test has limitations

because it requires a tilt table which may not be routinely available in all clinical settings. The sit-up test is used to overcome the limits of the HUT test, requires no equipment, and assesses postural changes in those who may have impaired balance or challenges with upright standing (Oyake et al., Blood Press Monit 2022, 27: 79-86). In patients with diabetes, OH can be correlated with increasing age, diabetic cardiac autonomic neuropathy, diabetes duration, uncontrolled diabetes, volume depletion, dehydration, and certain medications (Idiaquez Rios et al., J Diabetes Complications 2022, 36: 108169).

18:

A primary consideration for admitting or discharging a patient with diabetes (i.e., new-onset diabetes mellitus (DM) or pre-existing DM) is ensuring that the patient has the knowledge, tools, and caregiver support (if applicable) to effectively manage this condition at home, even in the absence of complicating factors. Adults who are socioeconomically disadvantaged, older, and/or uninsured, and patients with mental health conditions have significant risks and barriers that complicate outpatient management. Assessing for potential hazards and obstacles (e.g., inability to obtain medications and supplies, knowledge deficits related to blood glucose monitoring and insulin administration) is vital (Maloney and Glauser, In: Walls, ed. Rosen's Emergency Medicine: Concepts and Clinical Practice, 10th edn. 2022:1550-1). Safe and quality DM management and discharge planning involve assessing and identifying risks and barriers at admission and addressing them before discharge.

19:

The goals of diabetes treatment are to prevent and delay complications and enhance the patient's quality of life. Patient-centered glycemic management includes educating patients and their families or caregivers to promote informed and shared decision-making. A patient-centered focus considers the patient's preferences, chooses a regimen that optimizes tolerance and adherence, empowers the patient, and ensures diabetes self-management education and support access. Glycemic targets are a necessity to include in the patient-centered strategy. Simplifying the treatment plan can improve self-efficacy and the patient's ability to meet target goals. If needed, target goal adjustments are appropriate to improve safety and adherence to the treatment regimen (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Davies et al., Diabetes Care 2018, 41: 2669-701).

20:

Instruction: Medication adjustment includes initiation of a new drug, change in dosage, and/or administration frequency of a medication. Blood glucose values guide medication dosing decisions and adjustments. Administration of sliding scale insulin must occur at least 3 times within 24 hours. A continuous infusion rate must have at least 2 rate adjustments within 24 hours (including insulin pump rate adjustments). A glucose-lowering medication dose withheld based on normal or low blood glucose levels is included in the count of doses administered within the 24-hour timeframe.

21:

The target glycemic range for non-critically ill hospitalized patients with diabetes is 100-180 mg/dL (5.6-10.0 mmol/L). Blood glucose concentrations greater than 180 mg/dL (10.0 mmol/L) may be acceptable, at times, to reduce the risk of harmful outcomes in patients with severe comorbidities or decreased life expectancy. Blood glucose targets should be individualized and modified if there are identifiable risks for adverse effects such as a history of frequent hypoglycemia or hypoglycemia unawareness (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

22:

Scheduled subcutaneous insulin is recommended to manage hyperglycemia in non-critically ill hospitalized patients. The preferred treatment for patients with adequate nutritional intake includes an insulin regimen that consists of basal, prandial, and correction insulin aligned with meals. Recommendations are to complete point-of-care (POC) glucose testing and subcutaneous rapid or short-acting insulin administration immediately before meals. Alternatively, in patients with inadequate oral intake, prandial insulin administration is recommended directly after the patient eats, with adjustments in dosing related to the amount of food consumed. The preferred treatment for patients with oral intake restrictions includes basal insulin or basal plus correction insulin. For these patients or those on continuous enteral or parenteral nutrition, recommendations are to complete POC glucose testing and subcutaneous rapid or short-acting insulin administration every 4-6 hours (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

23:

When making treatment decisions, providers should assess for socioeconomic and psychosocial barriers and consider medication-taking behavior and the religious and cultural preferences of the patient. For non-critically ill patients with hyperglycemia and type 2 diabetes mellitus (DM) in the hospital, insulin is often the preferred treatment. However, based on patient preference, it can be appropriate to give oral glucose-lowering medication. If deemed appropriate by the provider, patients with preexisting DM may continue their home regimen, including oral glucose-lowering medication. When oral glucose-lowering medication is on hold during hospitalization, resumption of this medication may be suggested 1-2 days prior to discharge (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

24:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

25:

Patients with uncontrolled diabetes and symptomatic hyperglycemia are at risk for acute complications that necessitate fluid repletion; complications include, but are not limited to, dehydration. Dehydration is more likely to require hospitalization in individuals with diabetes than in those without diabetes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

26:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights. For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2. (Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

27:**Expected progress after 48-72h**

- Blood glucose stable or close to baseline
- Patient/caregiver knowledgeable and demonstrates self-administration of insulin with little or no assistance
- Adequate PO intake
- Activities at baseline
- Preparing for discharge within the next 24 hours

If no, consider:

- Endocrinology consultation for diabetic adjustment for uncontrolled diabetes mellitus or gastroparesis management
- Infectious disease consultation for persistent fever/infection
- Continued treatment of precipitating event

28:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

29:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

30:

A glucose target between 100-180 mg/dL (5.6-10.0 mmol/L) is recommended for most patients in noncritical care units (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

31:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

32:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

33:

National standards recommend diabetes self-management education and support (DSMES) for patients and/or caregivers to enhance knowledge, decision making, and mastery of skills for diabetes management to prevent acute complications, improve treatment outcomes, and reduce risk of long-term complications. Hospitalization represents a critical time to evaluate, assess, and provide DSMES that includes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Identifying a health care provider to provide diabetes care post discharge
- Establishing home blood glucose goals, self-monitoring of blood glucose, and understanding when to call the provider
- Recognition, treatment, and prevention of hyperglycemia and hypoglycemia
- Administration of blood glucose-lowering medications, including insulin, if relevant
- Information on healthy dietary choices
- Proper use and disposal of insulin needles and syringes
- Sick-day management (e.g., when sick with a cold, flu, or infection)

34:

Critical events such as a new diagnosis, unmet treatment goals, complications, and transitions in life require evaluation of diabetes self-management educational needs to encourage acquiring the skills that may assist in executing and adhering to the treatment regimen and medical nutrition therapy (MNT) plan. An individualized MNT plan created through a joint effort of shared decision making with the patient and the healthcare team that includes a registered dietitian nutritionist is essential. The goals of MNT are to attain individualized glycemic targets, delay or prevent complications, reach and maintain body weight goals, incorporate personal and cultural preferences, address barriers to behavioral changes and access to healthy foods, and provide the tools and resources to develop and maintain healthy dietary patterns. Individualized macronutrient recommendations consider the patient's eating patterns, preferences, and metabolic goals, as evidence shows there are no general recommendations for macronutrients for all people with diabetes. Carbohydrate monitoring strategies are essential to achieve glycemic control and include, but are not limited to, carbohydrate counting, an exchange system, or experience-based estimation (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; American Diabetes Association, Diabetes Care 2022, 45:S1-S262).

35:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

36:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

37:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

38:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

39:

Cardiovascular disease (CVD) is the major cause of mortality and morbidity in patients with diabetes mellitus (DM). A blood pressure level less than 130/80 mmHg is recommended in individuals with diabetes to reduce CVD mortality. Patients with DM should be educated about lifestyle modifications that effectively reduce hypertension and dyslipidemia. These modifications may include (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Weight loss, if overweight
- Dietary Approaches to Stop Hypertension (DASH), including reducing sodium and increasing potassium intake
- Reduction of saturated fat, trans fat, and cholesterol consumption
- Increase of n-3 fatty acids, viscous fiber, and plant stanols/sterols consumption
- Moderation of alcohol consumption
- Increase in physical activity

In addition to lifestyle therapy, statin therapy is recommended for those with established CVD disease and risk factors. Moderate-intensity statin therapy is recommended for individuals with diabetes aged 40-75 years without CVD. High-intensity statin therapy should be initiated in people with diabetes aged 40-75 with one or more CVD risk factors, to achieve an LDL cholesterol less than 70 mg/dL and reduce LDL cholesterol by 50%(0.50) from baseline. Treatment recommendations for individuals with diabetes and established CVD include high-intensity statin therapy to lower LDL cholesterol by 50%(0.50) from baseline and an LDL cholesterol goal of less than 55mg/dL. Individuals with diabetes age 75 and older already on statin therapy can continue statin treatment. It may also be reasonable to initiate moderate-intensity statin therapy in adults with diabetes aged 75 and older after discussing the potential benefits and risks (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

40:

Large prospective randomized studies have shown that intensive diabetes management to achieve near-normal glucose levels delays the onset of nephropathy and diabetic retinopathy. Improved glycemic control has also been shown to modestly slow the progression of neuropathy. Reducing hypertension is also an effective way of preventing the progression or development of nephropathy and retinopathy (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

41:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

42:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

43:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

44:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

45:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

46:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

47:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

48:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

49:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

50:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

51:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

52:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

53:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs

- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

54:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

55:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

56:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

57:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

58:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.